

Harshit Chelladurai

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EDUCATION

BMS College of Engineering, Bengaluru, India
Bachelor of Engineering, Information Science and Engineering

Jul 2018 – Aug 2022
CGPA – 9.18/10

TECHNICAL SKILLS

Languages: Python, Golang, Java, C++, JavaScript, SQL
Technical Skills: Object Oriented Programming, SOLID principles, Design Patterns, ACID
Frameworks: Flask, SprintBoot, ReactJS
Tools: Linux, Docker, Git, Github, MySQL, Postgresql, Anisble, Jenkins, Jira, Confluence, Drawio

WORK HISTORY

Akamai Technologies India Pvt Ltd.

Bengaluru, Karnataka

Software Engineer 2,

Jul 2024 – Present

Local Numa Aware Placement (Python):

- Improved system **maintainability** by applying the **Strategy Pattern** to decouple tightly integrated components, enabling seamless future **updates**.
- Reduced **intra-host VM migrations** by **60%** by designing and implementing a **resource allocation algorithm** to stabilize **system performance**.
- Optimized resource utilization** by developing a **VM packing algorithm** within **L3 cache boundaries**, facilitating efficient performance scaling and resource management.
- Increased **communication efficiency** by implementing a **gRPC-based server-client architecture** in **Python** for **low-latency** data exchange.

Software Engineer 1,

Jul 2022 – Jul 2024

Host Process Observability and Assurance (Python):

- Enhanced system **observability at scale** by developing a **process metrics scanner** deployed across **12,500+ hosts**, analyzing **4,000+ processes per host**, to deliver **real-time insights** into **system health and performance**.
- Ensured a successful **system rollout** by proactively identifying and resolving **critical bugs**, enabling **on-time production deployment** and minimizing **operational disruptions**.
- Detected **abnormal NGINX resource usage** across **30+ hosts**, leading to targeted **remediation and tuning** that optimized **resource consumption** and boosted **system efficiency**.

Pliny Aggregator (SpringBoot - Java):

- Enhanced system **observability** by developing **log aggregation services** that processed over **1M logs daily**, enabling better **monitoring and diagnostics**.
- Boosted **log processing performance** by optimizing **processing workflows**, resulting in a **33% reduction** in overall **processing time**.
- Ensured a **scalable and maintainable architecture** by applying **SOLID principles** in **Java Spring Boot**, improving long-term **code stability** and **developer productivity**.

PROJECTS

CLI Tools (Bash & Python):

- Streamlined the **developer workflow** by building a **tmux sessionizer tool** with **fzf**, simplifying **navigation** and **project session management**.
- Improved **personal productivity** by adding a **terminal-based timer** and integrating it into the **development workflow** using **Golang** and **tmux**.
- Built a **terminal-based journaling application** to enable fast, **distraction-free note-taking** with **CLI-first workflows**.
- Created a **command-line Wikipedia search tool**, leveraging **public APIs** to provide quick, in-terminal access to **reference content**.

Youtube Music Downloader (Python):

- Built an **offline music solution** that **downloaded 1,000+ tracks**, enabling uninterrupted playback without internet.
- Automated **audio fetching, conversion, and storage** by processing **playlist link inputs**, which reduced manual effort.